

more persuasive than the reality of uncertainty. So we cling to stories about outcomes being in our control.

Part of this has to do with confusing fields of precision with fields of uncertainty.

NASA's New Horizons spacecraft passed by Pluto two years ago. It was a three-billion mile trip that took nine and a half years. According to NASA, the trip "took about one minute less than predicted when the craft was launched in January 2006."⁶⁸

Think about that. In an untested, decade-long journey, NASA's forecast was 99.99998% accurate. That's like forecasting a trip from New York to Boston and being accurate to within four millionths of a second.

But astrophysics is a field of precision. It isn't impacted by the vagaries of human behavior and emotions, like finance is. Business, economics, and investing, are fields of uncertainty, overwhelmingly driven by decisions that can't easily be explained with clean formulas, like a trip to Pluto can. But we desperately want it to be like a trip to Pluto, because the idea of a NASA engineer being in 99.99998% control of an outcome is beautiful and comforting. It's so comforting that we're tempted to tell ourselves stories about how much control we have in other parts of our life, like money.

Kahneman once laid out the path these stories take:

- When planning we focus on what we want to do and can do, neglecting the plans and skills of others whose decisions might affect our outcomes.
- Both in explaining the past and in predicting the future, we focus on the causal role of skill and neglect the role of luck.
- We focus on what we know and neglect what we do not know, which makes us overly confident in our beliefs.

He described how this impacts businesses:

I have had several occasions to ask founders and participants in innovative start-ups a question: To what extent will the outcome of your effort depend on what you do in your firm? This is evidently an easy question; the answer comes quickly and it has never been less than 80%. Even when they are not sure they will succeed, these bold people think their fate is